

True or false? (unless otherwise specified.) Assume any explicitly named package is installed. An expression that raises an error does not evaluate to `true`.

1. The expression `"ab" * "cd" * "!"` returns the string `"abcd!"`.
2. The following code produces a single plot containing both curves, rather than replacing the first curve with the second.

```
using Plots
plot(x -> x^2, xlims=(-1, 1), label="square")
plot!(x -> x^3, label="cube")
```

3. The expression `eps()/2 + eps()/2 == eps()` evaluates to `true`.
4. The expression `typeof(1 + 2.0) == Float64` evaluates to `true`.
5. If `u = [1, 2]` and `v = [3, 4]`, then `[u v]` returns the matrix `[1 2; 3 4]`.
6. If `A = [1 2 3; 4 5 6]`, then `A[2, 3]` returns 6.
7. The following code assigns the vector `[2, 5]` to `w`.

```
v = [-3, 2, 0, 5, -1]
w = v[v .> 0]
```
8. The following code assigns the vector `[1, 3, 5]` to `a`.

```
f(x, y) = x - 2y
a = f.([3, 5, 7], 1)
```
9. The expression `sum(length, ["Julia", "is", "fun"])` returns 3.
10. The command `A = [i + 2j for i in 1:2, j in 1:3]` produces the matrix `A = [3 5 7; 4 6 8]`.

11. After the following definitions, `f(f(1))` returns 2.0.

```
f(x::Float64) = 1
f(x::Int) = 2.0
```

12. The call `odd_sum(5)` returns 9.

```
function odd_sum(n)
    s = 0
    for k in 1:n
        if isodd(k)
            s += k
        end
    end
    return s
end
```

13. If `x = LinRange(0, 1, 5)`, then the spacing between consecutive entries of `x` is 0.2.

14. The following code computes the first six Fibonacci numbers, and the final value of `f[6]` is 8. Recall that $f_1 = f_2 = 1$ and $f_k = f_{k-1} + f_{k-2}$ for $k \geq 3$.

```
f = [1, 1, 0, 0, 0, 0]
for k in 3:6
    f[k] = f[k-1] + f[k-2]
end
```

15. If `v = [2, 4, 6, 8]`, then `v[2:end]` returns the vector `[2, 4, 6]`.

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Bonus (2 points). Give the final values of `A` and `B` after the following code. Explain why modifying `B` and modifying `C` have different effects on `A`.

```
A = [1 2; 3 4]
B = A[:, 1]
C = view(A, :, 2)
B .= 0
C .= 9
```

Answer and explanation:

Answer key

Questions 1–15: 1 point for a correct answer, 0 otherwise.

1. **True.** `*` concatenates strings. The result is `"abcd!"`.
2. **True.** `plot!` adds a series to the current plot, which already contains the square function.
3. **True.** For `Float64`, `eps()` is 2^{-52} and `eps()/2` is 2^{-53} . Both are exactly representable, and adding the two halves gives `eps()` exactly.
4. **True.** The integer is promoted for this addition; the result is `3.0`, of type `Float64`.
5. **False.** Horizontal concatenation places `u` and `v` in the columns, giving `[1 3; 2 4]`.
6. **True.** The first index selects the row and the second selects the column. The entry in row 2, column 3 is 6.
7. **True.** The mask selects the strictly positive entries, giving `[2, 5]`.
8. **True.** The scalar second argument is used in each call: `f(3, 1)`, `f(5, 1)`, and `f(7, 1)`.
9. **False.** The expression sums the string lengths, giving $5 + 2 + 3 = 10$, rather than counting the strings.
10. **True.** Evaluating `i + 2j` for each row index `i` and column index `j` gives `[3 5 7; 4 6 8]`.
11. **False.** The inner call `f(1)` returns `2.0`. The outer call therefore uses the `Float64` method and returns 1.
12. **True.** The accumulator adds the odd integers $1 + 3 + 5 = 9$.
13. **False.** `LinRange(0, 1, 5)` has five equally spaced entries, including both endpoints. There are four intervals, so the spacing is $(1 - 0)/(5 - 1) = 0.25$.
14. **True.** Each new entry is the sum of the previous two. The final vector is `[1, 1, 2, 3, 5, 8]`.
15. **False.** `end` denotes the last index, which is 4. The slice selects entries 2 through 4, giving `[4, 6, 8]`.

Bonus. `A = [1 9; 3 9]` and `B = [0, 0]`. Extracting `A[:, 1]` creates a copy, so modifying `B` leaves the first column of `A` unchanged. In contrast, `C` is a view into the second column of `A`, so `C .= 9` modifies that column in the original matrix. Award 0.5 points for each correct final value and 0.5 points for each correct explanation (copy versus view).